Adithya Chavali

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Education

Indian Institute of Information Technology, Allahabad Aug 2023 – May 2027
B.Tech in Electronics and Communication Engineering *Prayagraj, India*

Relevant Coursework

Data Structures & Algorithms • Database Management Systems • Operating Systems • Computer Networks • AI and ML Techniques

Experience

Velind Virtual Systems Pvt. Ltd. June 2026 – July 2026
Software Development Intern *Hyderabad, India*

- Owned the permission layer as integration lead on a 5-person team, building role-based access control (RBAC) for 12 user roles and 20+ permission types and extending it as new modules were added.
- Wrote and ran 13 Alembic migrations as the schema grew from 6 to 10 modules, and diagnosed and resolved a migration-chain conflict between two independently-developed changes.
- Built a project-detail aggregation feature pulling data from six modules into a single consolidated view, plus a real-time notifications system with unread tracking.
- Worked on an executive dashboard (React, Recharts) with live KPI cards and charts, and caught two integration bugs during testing before they reached production.

Ada Insys Pvt. Ltd. May 2025 – July 2025
Software Development Intern *Hyderabad, India*

- Led a 3-person team building a biometric attendance platform (Node.js, PostgreSQL) with rotating QR codes and facial recognition, supporting 1,000+ student sessions.
- Cut proxy attendance by 90% with time-bound QR validation algorithms and Redis caching for real-time code generation.
- Built out normalized PostgreSQL schemas and REST APIs across 3 core modules (authentication, attendance tracking, reporting), tested with Postman before rollout.

Projects

DevLog | *TypeScript, Node.js/Express, SQL, PostgreSQL, Redis, BullMQ, Terraform, AWS* GitHub

- Built a full-stack developer-activity platform: a React dashboard and REST API backend ingesting GitHub commit diffs via webhooks into a normalized relational schema (PostgreSQL/Prisma).
- Built a Redis + BullMQ ingestion pipeline with exponential-backoff retries and database-level idempotency, plus AES-256-GCM encrypted token storage and Terraform-provisioned AWS infrastructure.

Watchdog | *Go, React, TypeScript, SQL, PostgreSQL, Prometheus, Docker* GitHub

- Built a concurrent endpoint-monitoring engine in Go with a bounded worker pool, per-monitor schedulers, and a deterministic health state machine, verified race-free with Go's race detector.
- Benchmarked the worker pool to 22K+ jobs/sec at 50 workers, containerized with Docker, and gated every push with a CI test suite and race detector.

Conquer | View Project | *Next.js, TypeScript, Drizzle ORM, SQL, PostgreSQL (Neon), Groq, Gemini, OpenAI* GitHub

- Built a full-stack interview simulator with a real-time streaming chat UI in Next.js/TypeScript, backed by a multi-provider LLM gateway (Groq, Gemini, OpenAI) that automatically switches providers if one goes down.
- Built an agent that routes follow-up questions through a concept graph, blending LLM judgment with simple rule checks instead of relying on the LLM for everything.

Benizakura | *Python, pytest, Statistical Analysis, CI/CD* GitHub

- Built a Python framework that evaluates AI outputs pairwise and bidirectionally, writing clear, well-tested code so disagreements are flagged as unreliable rather than silently trusted.
- Built the core evaluation engine with JSON Schema-validated output contracts and 198 passing pytest tests, using Cohen's κ agreement statistics for human calibration, gated by CI before every merge.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, Go, SQL, C, C++, HTML/CSS, Bash
Frontend: React.js, Next.js, Vite, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Prisma, Drizzle ORM
Databases: PostgreSQL, MongoDB (NoSQL), Redis, Supabase, Convex, Neon
Cloud & Infrastructure: AWS (EC2, VPC, IAM, Security Groups), Docker, Terraform
Testing & Tools: pytest, Vitest, Git, GitHub Actions (CI/CD), Postman, Playwright
AI & LLM Integration: Groq API, Gemini API, OpenAI API, RAG pipelines, multi-provider LLM gateways, agentic workflows